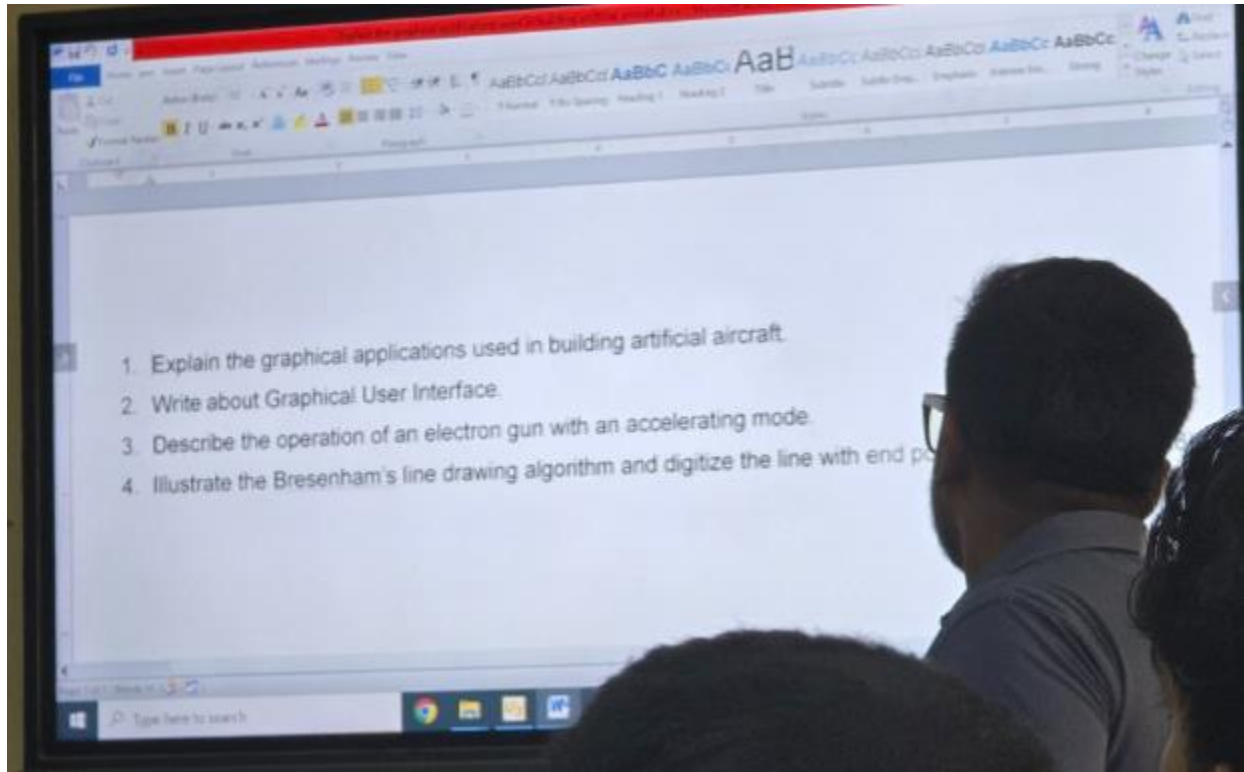


ALL CT & ASSIGNMENT QUESTION CSE 4-1 SESSION:21-22

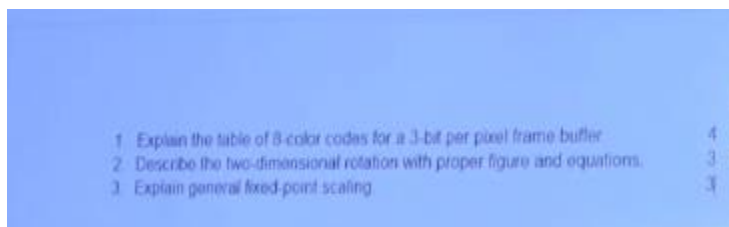
CSE-4101 Computer Graphics and Multimedia

CT-01



- 1.Explain the graphical applications used in building artificial aircraft.
- 2.Write about Graphical User Interface.
- 3.Describe the operation of an electron gun with an accelerating mode.
- 4.Illustrate the Bresenham's Line Drawing algorithm and digitize the line with end point (20,10) and (30,18)

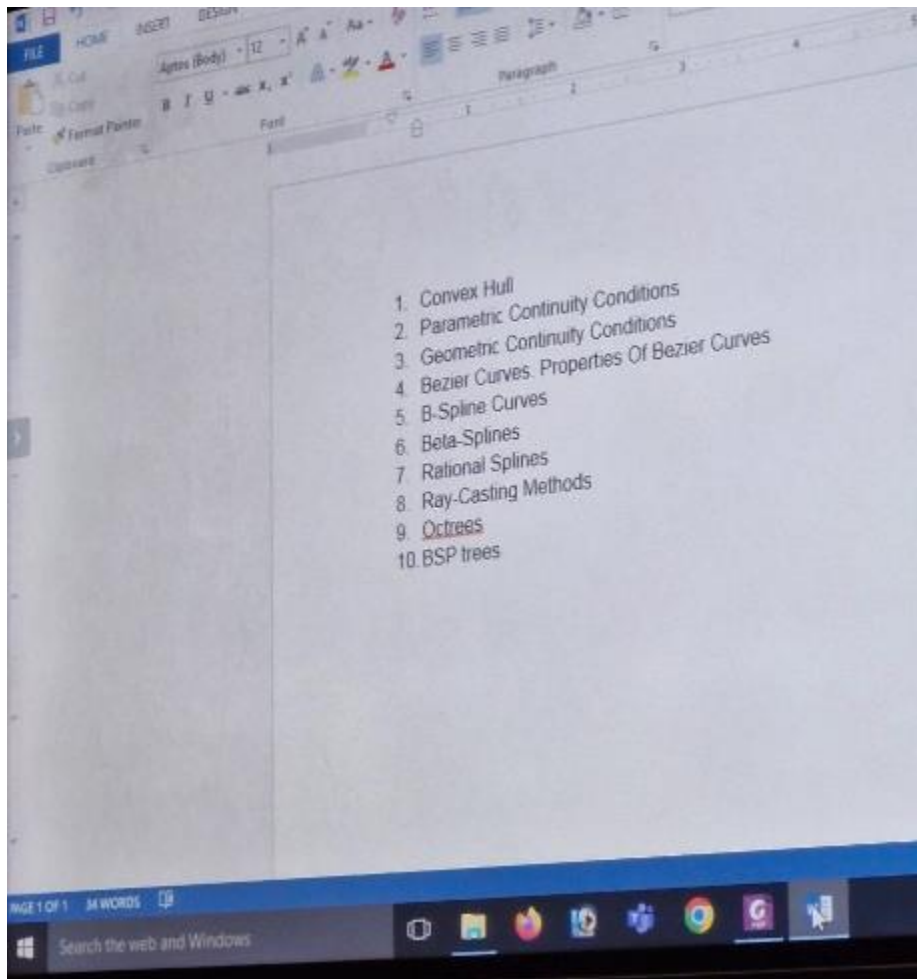
CT-02



1. Explain the table of 8-color codes for a 3 bit per pixel frame buffer.
2. Describe the two dimensional rotation with proper figure and equations.

3. Explain general fixed point scaling.

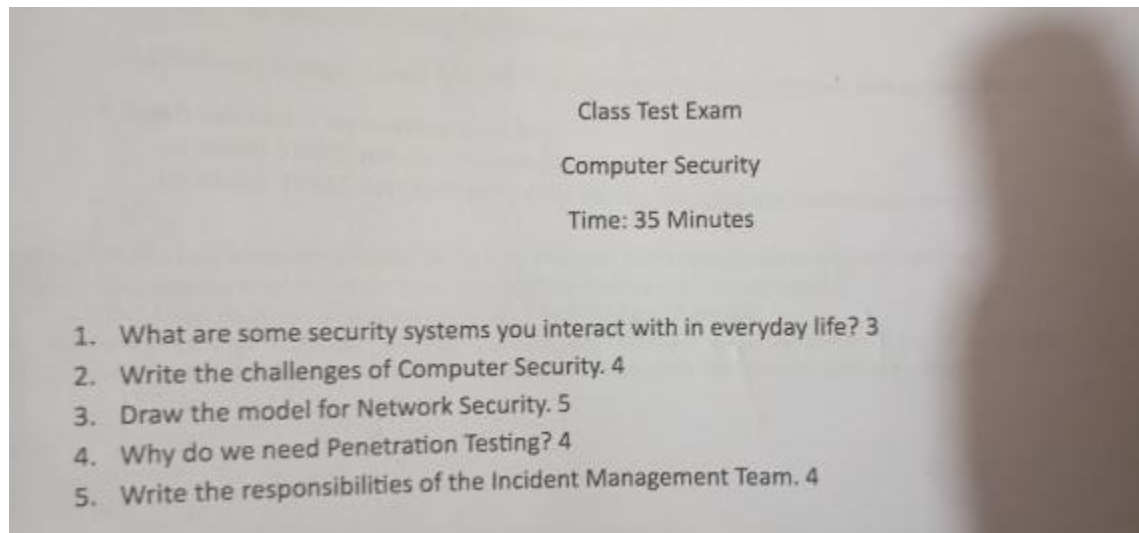
Assignment-01



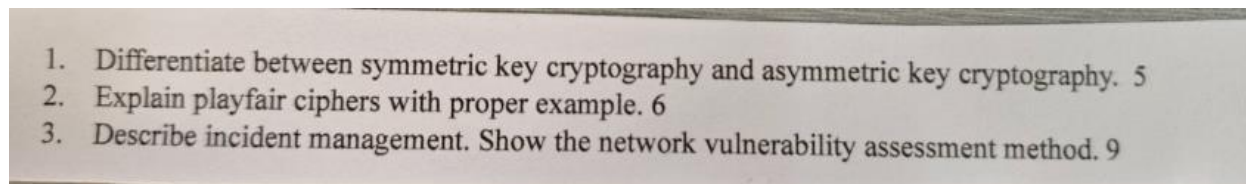
Assignment extends:

1. Translation, rotation and scaling in three-dimensional transformation. 2. 3D object representation. 3. List and definitions of 3D object representation categories. 4. Perspective projection matrix. 5. Explanations of equations for perspective projection matrix. 6. General parallel projection transformation. 7. Briefly explain the operations of clipping. 8. Bezier Curves and properties of Bezier Curves. 9. Steps for depth-buffer algorithm. 10. Basic functions of depth-sorting method. Graphics Extra Topic assignment add hobe r agar sob topic lagbe Suggestion all topic

CT-01



CT-02



ASSIGNMENT-01

1. What is the primary purpose of the private key in RSA? Given $(p=19)$, $(q=23)$, and public exponent $(e=5)$, generate the RSA keys and encrypt the message $(M=42)$, then verify the original message by decrypting the ciphertext. (4)
2. Given the public prime $(p = 23)$ and generator $(g = 5)$, Alice chooses a private key $(a = 6)$ and Bob chooses a private key $(b = 15)$. Compute Alice's and Bob's public keys, and determine the shared secret key established between them. (3)
3. Encrypt the plaintext "COMPUTERNETWORKS" using a 3-rail Rail Fence Cipher, then decrypt the resulting ciphertext to recover the original message. (3)

BUS-4101 E COMMERCE AND MANAGEMENT INFORMATION SYSTEM

CT-01

1. An e-commerce company's online sales are increasing, but profits are decreasing due to high marketing and delivery costs. Analyze the situation using: (5)
 - Value Chain, CAC, CLV, Outbound Logistics; (Suggest suitable improvements.)
2. Explain the concepts of Disintermediation and Re-intermediation in E-Commerce. (5)
How can both occur simultaneously in the same industry? Use suitable real-world examples.
3. Daraz experiences a huge traffic spike during its "11.11 Campaign." Millions of users browse products at the same time, causing high server load. (5)
Analyze how the following technologies help Daraz maintain performance and availability during the sale: (Also explain what problems may occur if these technologies are not used.)
 - CDN, Load Balancer, Horizontal Scaling, Database Read Replicas
4. Why network effects are important for digital platforms? Calculate using Metcalfe's Law, if a platform has 15 users and later grows to 30 users, by what factor does the network value increase? (1+4)

CT-02

1. Define MIS. Distinguish MIS from a Data Processing System. (3)
2. Using Anthony's Triangle, explain how TPS, MIS, DSS, and EIS support different management levels. (3)
3. A Bangladeshi bank is implementing a new MIS. (4)
 - (a) Identify THREE potential risks during/after implementation.
 - (b) Identify THREE Critical Success Factors that will determine implementation success.

ASSIGNMENT -01

A PROJECT ON E-COMMERCE WEBSITE SHOULD BE DONE BY EVERY SINGLE STUDENT

CSE-4115

SOFTWARE DESIGN PATTERN AND PRINCIPLES

CT-01

1.Explain Qualified Association with proper class diagram and pseudocode -20

ASSIGNMENT -01

1.Show Abstract Factory Design Code on C#.-20

ASSIGNMENT-02

1.Show Decorator Design Code on C#.-20